The Derivative from First Principles

Section 1: The Limit Definition of the Derivative

1 State the standard limit definition of $f'(x)$.

a. $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

b. $f'(x) = \lim_{h \to 0} \frac{f(x) - f(h)}{h}$

c. $f'(x) = \frac{f(x+h) - f(x)}{h}$

d. $f'(x) = \lim_{x \to 0} \frac{f(x+h) - f(x)}{x}$

2 Use first principles to find the derivative of $f(x) = 5x^2 - 3x$.

a. $10x - 3$

b. $10x$

c. $5x - 3$

d. $10x + 3$

3 Use the alternative limit definition $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3$.

a. 12

b. 8

c. 6

d. 4

Section 2: First Principles for Radicals and Reciprocals

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4 Find $f'(x)$ from first principles for $f(x) = \frac{1}{x}$.

a. $-1/x^2$

b. $1/x^2$

c. $-1/x$

d. $-2/x^2$

5 Find $f'(x)$ from first principles for $f(x) = \sqrt{x}$ $(x > 0)$.

a. $\frac{1}{2\sqrt{x}}$

b. $\frac{1}{\sqrt{x}}$

c. $-\frac{1}{2\sqrt{x}}$

d. $\frac{2}{\sqrt{x}}$

6 Evaluate $\lim_{h \to 0} \frac{(2+h)^5 - 32}{h}$ by recognizing it as a derivative.

a. 80

b. 40

c. 32

d. 16

Section 3: Differentiability and Continuity

7 Which statement correctly describes the relationship between differentiability and continuity?

a. Differentiability implies continuity, but continuity does not imply differentiability

b. Continuity implies differentiability

c. They are completely independent properties

d. A function is differentiable if and only if it is continuous

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8 Why does $f(x) = |x - 3|$ fail to be differentiable at $x = 3$?

a. The left-hand and right-hand derivative limits are unequal ($-1 \neq 1$)

b. $f(3)$ is undefined

c. f is discontinuous at $x = 3$

d. The tangent line is horizontal

9 At $x = 0$, the function $f(x) = x^{1/3}$ has which type of non-differentiability?

a. Vertical tangent ($f'(x) \to \infty$)

b. Corner / sharp turn

c. Cusp

d. Removable discontinuity

Section 4: Geometric and Physical Interpretations

10 What geometric feature does the derivative $f'(a)$ represent?

a. The slope of the tangent line to $y = f(x)$ at $x = a$

b. The area under the curve from 0 to a

c. The secant line slope between 0 and a

d. The curvature at $(a, f(a))$

11 A particle moves along a line with position $s(t) = 4t^2 - 3t + 1$. What is its instantaneous velocity at $t = 2$?

a. 13

b. 11

c. 16

d. 10

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12 Find the equation of the tangent line to $y = x^2 - 4x$ at the point where $x = 3$.

a. $y = 2x - 9$

b. $y = 2x - 3$

c. $y = -x$

d. $y = 4x - 15$

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